

2024-09-24-SimplexMatrix

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Slide 1

Content

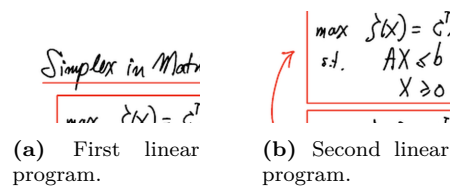
Simplex in Matrix Notation

$$\begin{pmatrix} \max & f(x) = c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0 \end{pmatrix} \quad x \in \mathbb{R}^n, c \in \mathbb{R}^n$$
$$A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$$
$$\begin{pmatrix} \max & f(x) = c^T x \\ \text{s.t.} & W = b - Ax \\ & x, W \geq 0 \end{pmatrix} \quad W \in \mathbb{R}^m$$

Description

This slide introduces two equivalent formulations of a linear program in matrix notation, relevant to a Linear Programming course. The top formulation is a standard linear program: maximize $f(x) = c^T x$ subject to $Ax \leq b$ and $x \geq 0$. The variables are defined: $x \in \mathbb{R}^n$ is the decision variable vector, $c \in \mathbb{R}^n$ is the cost vector, $A \in \mathbb{R}^{m \times n}$ is the constraint matrix, and $b \in \mathbb{R}^m$ is the right-hand side vector. The bottom formulation introduces slack variables $W \in \mathbb{R}^m$, transforming the inequality constraints into equality constraints: $W = b - Ax$, with the additional non-negativity constraint $W \geq 0$. Both formulations aim to solve the same optimization problem, but the second one is often preferred for algorithmic purposes (like the simplex method) because it deals with equality constraints. The curved arrow indicates the equivalence between the two formulations. The context of the slide is clearly within the topic of linear programming, specifically showing how to represent a linear program using matrix notation, a crucial step for applying algorithmic solutions.

Figures



Slide 2

Content

Simplex in Matrix Notation

$$\begin{pmatrix} \max & f(x) = c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0 \end{pmatrix} \quad x \in \mathbb{R}^n, c \in \mathbb{R}^n$$

$$A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$$

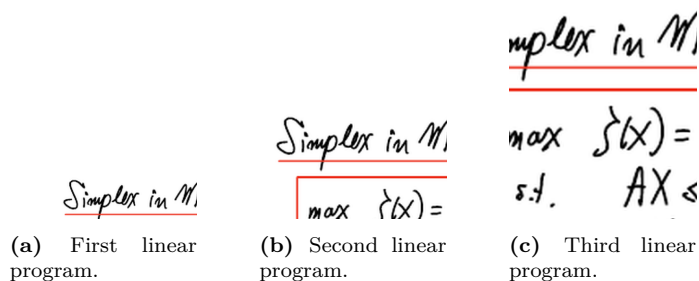
$$\begin{pmatrix} \max & f(x) = c^T x \\ \text{s.t.} & W = b - Ax \\ & x, W \geq 0 \end{pmatrix} \quad W \in \mathbb{R}^m$$

$$\begin{pmatrix} \max & f(x) = \bar{c}^T \bar{x} \\ \text{s.t.} & \bar{W} = \bar{b} - A\bar{x} \\ & \bar{x}, \bar{W} \geq 0 \end{pmatrix}$$

Description

This slide illustrates three equivalent formulations of a linear program in matrix notation, crucial for understanding the simplex method in a Linear Programming course. The first formulation presents a standard maximization linear program: maximize $f(x) = c^T x$ subject to $Ax \leq b$ and $x \geq 0$, where $x \in \mathbb{R}^n$ is the decision variable vector, $c \in \mathbb{R}^n$ is the cost vector, $A \in \mathbb{R}^{m \times n}$ is the constraint matrix, and $b \in \mathbb{R}^m$ is the right-hand side vector. The second formulation introduces slack variables $W \in \mathbb{R}^m$, converting inequality constraints to equality constraints: $W = b - Ax$, with $x, W \geq 0$. The third formulation represents an intermediate step in the simplex algorithm, where the objective function is updated to $f(x) = \bar{c}^T \bar{x}$, and the constraints are similarly modified to $\bar{W} = \bar{b} - A\bar{x}$, with $\bar{x}, \bar{W} \geq 0$. The curved arrows indicate the iterative process of the simplex method, transforming the initial linear program into the final form. The notation $\bar{x}$, $\bar{W}$, $\bar{b}$, and $\bar{c}$ suggests that these are updated values during the simplex iterations. The slide highlights the transition from the initial problem to the form used in the simplex algorithm.

Figures



Slide 3

Content

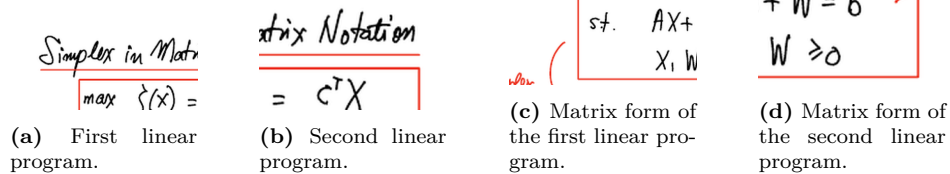
Simplex in Matrix Notation

$$\left(\begin{array}{ll} \max & \xi(x) = c^T x \\ \text{s.t.} & Ax + W = b \\ & x, W \geq 0 \end{array} \right) \xrightarrow{\text{Simplex iterates}} \left(\begin{array}{ll} \max & \xi(x) = \bar{\beta} + \bar{c}^T x \\ \text{s.t.} & Ax + W = b \\ & x, W \geq 0 \end{array} \right)$$

Description

This slide, from a Linear Programming course, demonstrates how the simplex method iteratively transforms a linear program. The top box shows the initial linear program in standard form: maximize $\xi(x) = c^T x$ subject to $Ax + W = b$ and $x, W \geq 0$, where x are the decision variables, W are slack variables, c is the cost vector, A is the constraint matrix, and b is the right-hand side vector. The bottom box shows the same linear program after one or more iterations of the simplex method. The objective function is now expressed as $\xi(x) = \bar{\beta} + \bar{c}^T x$, where $\bar{\beta}$ represents the current objective function value and $\bar{c}$ is the updated cost vector. The constraints remain the same. The red arrow indicates the iterative nature of the simplex method, transforming the initial linear program into a modified form during each iteration. The matrix representations to the right show how the linear programs can be written in matrix form, which is crucial for implementing the simplex algorithm. The dimensions of the matrices are also indicated. The context of the slide is clearly within the simplex method of linear programming, showing the iterative process of transforming the problem into a form suitable for solution.

Figures



Slide 4

Content

Basic and Non-Basic Variables

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ w_1 \\ \vdots \\ w_m \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \end{pmatrix} = \begin{pmatrix} X_N \\ X_B \end{pmatrix} \quad N - \text{non-basic}$$

B - basic

$$\begin{aligned} c^T X &= \sum_{j=1}^n c_j x_j = \sum_{j=1}^n c_j x_j + \sum_{j=n+1}^{n+m} 0 \cdot x_j \\ &= \sum_{j \in B} c_j x_j + \sum_{j \in N} c_j x_j \\ &= c_B^T X_B + c_N^T X_N \end{aligned}$$

Description

This slide from a Linear Programming course explains the concept of basic and non-basic variables within the context of the simplex method. The slide begins by partitioning the vector of variables X into two parts: X_B (basic variables) and X_N (non-basic variables). The initial vector X contains decision variables x_i and slack variables w_i . The equation $c^T X = \sum_{j=1}^n c_j x_j$ represents the objective function, where c_j are the coefficients and x_j are the variables. The slide then shows how this objective function can be rewritten by separating the basic and non-basic variables. The terms $\sum_{j=n+1}^{n+m} 0 \cdot x_j$ represent the contribution of the non-basic variables to the objective function in the initial tableau (where they are set to zero). The final expression $c_B^T X_B + c_N^T X_N$ shows the objective function expressed in terms of basic and non-basic variables, which is a crucial step in the simplex algorithm. The curved arrows indicate the iterative nature of the simplex method, where variables switch between basic and non-basic status during iterations. The notation B and N represent the index sets of basic and non-basic variables, respectively. The context of the slide is clearly within the simplex method, specifically focusing on how the variables are partitioned and how this partitioning affects the objective function.

Figures

Basic and 1

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ w_1 \\ \vdots \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \end{pmatrix}$$

(a) Partitioning of the variable vector X .

Slide 5

Content

Basic and Non-Basic Variables

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ w_1 \\ \vdots \\ w_m \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \end{pmatrix} = \begin{pmatrix} X_N \\ X_B \end{pmatrix} \quad N - \text{non-basic}$$

B - basic

$$AX + W = b \iff \sum_{j=1}^n a_{ij}x_j + x_{n+i} = b_i$$

$$\begin{bmatrix} A & I \end{bmatrix} \begin{bmatrix} X \\ W \end{bmatrix} = b \iff \sum_{j=1}^{n+m} a_{ij}x_j = b_i$$

$$\sum_{j \in B} a_{ij}x_j + \sum_{j \in N} a_{ij}x_j = b_i$$

Description

This slide from a Linear Programming course introduces the concepts of basic and non-basic variables in the context of the simplex method. The slide begins by partitioning the vector of variables X , which includes decision variables x_i and slack variables w_i , into basic variables (X_B) and non-basic variables (X_N). The constraints of the linear program are represented as $AX + W = b$, where A is the constraint matrix, and b is the right-hand side vector. The slide then shows how this equation can be rewritten using the augmented

matrix $\begin{bmatrix} A & I \end{bmatrix}$, where I is the identity matrix. The equation $\sum_{j=1}^{n+m} a_{ij}x_j = b_i$ represents the i -th constraint

in terms of all variables. The slide further decomposes this equation into a sum over basic variables ($j \in B$) and non-basic variables ($j \in N$). The red curved arrow indicates the iterative process of the simplex method, where variables switch between basic and non-basic status during iterations. The notation B and N represent the index sets of basic and non-basic variables, respectively. The new matrix A is of dimension $m \times (n + m)$. The context of the slide is clearly within the simplex method, specifically focusing on how the variables are partitioned and how this partitioning affects the constraint equations.

Figures

Basic

(a) Partitioning of the variable vector X .

$V = \begin{pmatrix} x_1 \\ \vdots \end{pmatrix}$

(b) Augmented matrix $\begin{bmatrix} A & I \end{bmatrix}$.

Slide 6

Content

Basic and Non-Basic Variables

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ w_1 \\ \vdots \\ w_m \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \end{pmatrix} = \begin{pmatrix} X_N \\ X_B \end{pmatrix} \quad N - \text{non-basic}$$

B - basic

$$AX + W = b \iff \sum_{j=1}^n a_{ij}x_j + x_{n+i} = b_i$$

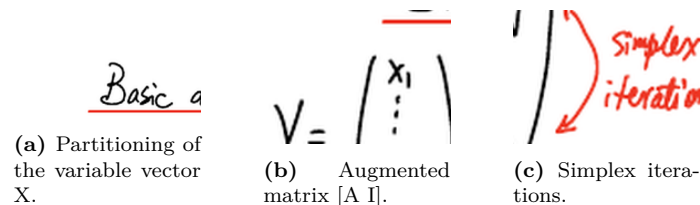
$$\begin{bmatrix} A & I \end{bmatrix} \begin{pmatrix} X \\ W \end{pmatrix} = b \iff \sum_{j \in B} a_{ij}x_j + \sum_{j \in N} a_{ij}x_j = b_i$$

$$BX_B + NX_N = b$$

Description

This slide, from a Linear Programming course, further develops the concepts of basic and non-basic variables introduced in the previous slide. It begins by partitioning the variable vector X , which contains both decision variables (x_i) and slack variables (w_i), into basic variables (X_B) and non-basic variables (X_N). The constraints of the linear program are represented as $AX + W = b$. The slide then shows how this equation can be rewritten using an augmented matrix $\begin{bmatrix} A & I \end{bmatrix}$, where I is the identity matrix. The equation is then decomposed into a sum over basic variables ($j \in B$) and non-basic variables ($j \in N$). The final equation, $BX_B + NX_N = b$, concisely represents the constraints in terms of basic and non-basic variables. The red curved arrow indicates the iterative nature of the simplex method, where variables change between basic and non-basic status during iterations. The notation B and N represent the index sets of basic and non-basic variables, respectively. The new matrix A is of dimension $m \times (n + m)$. The context of the slide is clearly within the simplex method, specifically focusing on how the variables are partitioned and how this partitioning affects the constraint equations. The slide builds upon the previous slide by showing how the matrix representation of the constraints is modified during the simplex iterations.

Figures



Slide 7

Content

Basic and Non-Basic Variables

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ w_1 \\ \vdots \\ w_m \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \end{pmatrix} = \begin{pmatrix} X_N \\ X_B \end{pmatrix} \quad N - \text{non-basic}$$

B - basic

$$[A \quad I] \begin{bmatrix} X \\ W \end{bmatrix} = b \iff [B \quad N] \begin{bmatrix} X_B \\ X_N \end{bmatrix} = b$$

B = basic var. cols of $[A \quad I]$, N = non-basic var. cols of $[A \quad I]$

Description

This slide from a Linear Programming course explains the concept of basic and non-basic variables within the context of the simplex method. The slide begins by partitioning the vector of variables X , which includes decision variables x_i and slack variables w_i , into basic variables (X_B) and non-basic variables (X_N). The constraints of the linear program are represented as $[A \quad I] \begin{bmatrix} X \\ W \end{bmatrix} = b$, where A is the constraint matrix, I is the identity matrix, and b is the right-hand side vector. The slide then shows how this equation can be rewritten as $[B \quad N] \begin{bmatrix} X_B \\ X_N \end{bmatrix} = b$, where B represents the columns of $[A \quad I]$ corresponding to the basic variables, and N represents the columns corresponding to the non-basic variables. The red curved arrow indicates the iterative nature of the simplex method, where variables switch between basic and non-basic status during iterations. The notation B and N represent the index sets of basic and non-basic variables, respectively. The context of the slide is clearly within the simplex method, specifically focusing on how the variables are partitioned and how this partitioning affects the constraint equations. The slide builds upon the previous slide by showing how the matrix representation of the constraints is modified during the simplex iterations.

Figures

Basic

(a) Partitioning of the variable vector X .

$V = \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ w_1 \\ \vdots \\ w_m \end{pmatrix}$

(b) Augmented matrix $[A \quad I]$.

$= \begin{pmatrix} \cdot \\ \cdot \\ x_n \\ w_1 \\ \cdot \\ \cdot \end{pmatrix} =$

(c) Simplex iterations.

Slide 8

Content

Simplex Iterations in Matrix Form

$$\begin{aligned}
 (1) \quad & BX_B + NX_N = b \implies X_B = B^{-1}b - B^{-1}NX_N \\
 (2) \quad & J = C_B^T X_B + C_N^T X_N \\
 &= C_B^T (B^{-1}b - B^{-1}NX_N) + C_N^T X_N \\
 &= C_B^T B^{-1}b - (C_B^T B^{-1}N - C_N^T)X_N \\
 &= \bar{J} + \bar{C}^T X_N
 \end{aligned}$$

Description

This slide from a Linear Programming course demonstrates the simplex iterations in matrix form. The slide is divided into two main parts. The first part (1) shows how to express the basic variables (X_B) in terms of the non-basic variables (X_N) using matrix notation. The equation $BX_B + NX_N = b$ represents the constraints in matrix form, where B and N are matrices formed from the columns of the augmented matrix corresponding to basic and non-basic variables, respectively. Solving for X_B yields $X_B = B^{-1}b - B^{-1}NX_N$. The second part (2) shows how to express the objective function (J) in terms of the non-basic variables. Starting with $J = C_B^T X_B + C_N^T X_N$, where C_B and C_N are vectors of coefficients corresponding to basic and non-basic variables, the expression is simplified by substituting the expression for X_B from part (1). Through algebraic manipulation, the objective function is rewritten as $J = \bar{J} + \bar{C}^T X_N$, where $\bar{J}$ is a constant and $\bar{C}$ is a vector representing the reduced costs. The red boxes highlight key results. The superscript T denotes the transpose of a matrix or vector. The context of the slide is clearly within the simplex method, specifically showing how matrix algebra is used to efficiently update the solution during iterations.

Figures

<u>Simplex Iterations in Matrix Form</u>	
(a) Equation 1: Matrix form of simplex iteration.	(b) Equation 2: Derivation of ob- jective function in matrix form.

Slide 9

Content

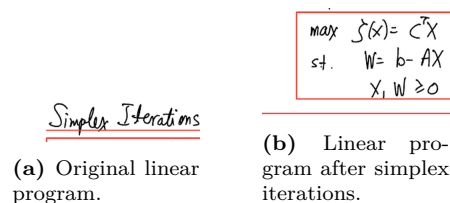
Simplex Iterations in Matrix Form

$$\begin{aligned}
 \max \quad & \mathcal{J}(x) = c^T x \quad x \in \mathbb{R}^n, w \in \mathbb{R}^m \\
 \text{s.t.} \quad & w = b - Ax \\
 & x, w \geq 0 \quad \implies \text{Simplex iterations} \\
 \max \quad & \mathcal{J}(x) = c_B^T B^{-1}b - ((B^{-1}N)^T c_B - c_N)^T x_N \\
 \text{s.t.} \quad & x_B = B^{-1}b - B^{-1}Nx_N \\
 & x_B, x_N \geq 0 \\
 & B, N \text{ are basic and non-basic var. cols of } [A \ I]
 \end{aligned}$$

Description

This slide, from a Linear Programming course, presents the simplex method in matrix form. The top section shows the original linear program: maximize the objective function $\mathcal{J}(x) = c^T x$ subject to the constraints $w = b - Ax$ and $x, w \geq 0$, where x represents the decision variables, w represents the slack variables, c is the cost vector, A is the constraint matrix, and b is the right-hand side vector. The variables x and w are in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively. A curved arrow points to the phrase "Simplex iterations," indicating that the following equations represent the linear program after applying simplex iterations. The bottom section shows the transformed linear program after applying simplex iterations. The objective function is now expressed as $\mathcal{J}(x) = c_B^T B^{-1}b - ((B^{-1}N)^T c_B - c_N)^T x_N$, and the constraints are $x_B = B^{-1}b - B^{-1}Nx_N$ and $x_B, x_N \geq 0$. Here, B and N are matrices formed from the columns of the augmented matrix $[A \ I]$ corresponding to basic and non-basic variables, respectively; c_B and c_N are the cost vectors corresponding to basic and non-basic variables. The final line explains that B and N represent the basic and non-basic variable columns of the augmented matrix $[A \ I]$. The slide demonstrates how the simplex method transforms the original linear program into a form suitable for iterative solution.

Figures



Slide 10

Content

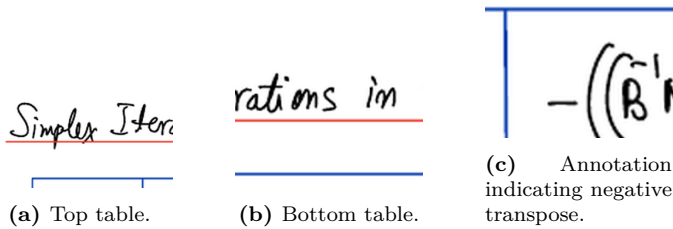
Simplex Iterations in Matrix Form

$c_B^T B^{-1} b$	$-((B^{-1} N)^T c_B - c_N)^T$	$-c_B^T B^{-1} b$	$-(B^{-1} b)^T$
$B^{-1} b$	$-B^{-1} N$	$(B^{-1} N)^T c_B - c_N$	$(B^{-1} N)^T$

Description

This slide from a Linear Programming course presents two tables summarizing the matrix form of simplex iterations. The top table (labeled (P)) shows the updated objective function value ($c_B^T B^{-1} b$) and the updated basic variable values ($B^{-1} b$) along with their respective changes due to changes in non-basic variables ($-((B^{-1} N)^T c_B - c_N)^T$ and $-B^{-1} N$). The bottom table (labeled (D)) appears to represent the dual simplex method, showing the negative of the objective function value ($-c_B^T B^{-1} b$) and the negative transpose of the basic variable values ($-(B^{-1} b)^T$). The other entries ($(B^{-1} N)^T c_B - c_N$ and $(B^{-1} N)^T$) likely represent the changes in the dual variables. A red curved arrow points to the bottom right entry of the top table and is annotated "negative transpose," indicating that the matrix is obtained by taking the negative transpose of another matrix. The notation B , N , c_B , c_N , and b are consistent with the previous slides, representing the basic variable matrix, non-basic variable matrix, basic variable cost vector, non-basic variable cost vector, and the right-hand side vector, respectively. The context of the slide is clearly within the simplex method, specifically showing the matrix operations involved in updating the solution and objective function during iterations. The two tables likely represent the primal and dual simplex methods, respectively.

Figures



Slide 11

Content

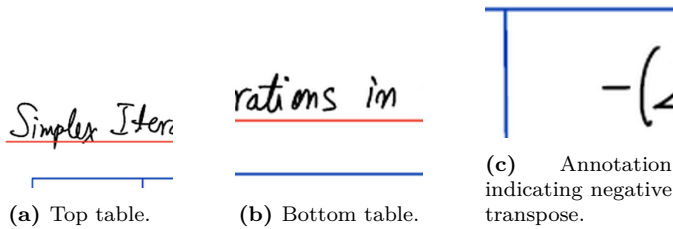
Simplex Iterations in Matrix Form

$\mathcal{J}^*$	$-(\mathcal{Z}_N^*)^T$	$-\mathcal{J}^*$	$-(\mathcal{X}_B^*)^T$
$\mathcal{X}_B^*$	$-\mathcal{B}^{-1}\mathcal{N}$	$\mathcal{Z}_N^*$	$(\mathcal{B}^{-1}\mathcal{N})^T$

Description

This slide from a Linear Programming course presents two tables summarizing the matrix form of simplex iterations. The top table, labeled (P), likely represents the primal simplex method. It shows the optimal objective function value ($\mathcal{J}^*$), the optimal values of the basic variables ($\mathcal{X}_B^*$), and how these values change with changes in the non-basic variables ($-(\mathcal{Z}_N^*)^T$ and $-\mathcal{B}^{-1}\mathcal{N}$). The bottom table, labeled (D), likely represents the dual simplex method. It shows the negative of the optimal objective function value ($-\mathcal{J}^*$), the negative transpose of the optimal basic variable values ($-(\mathcal{X}_B^*)^T$), and the changes in the dual variables ($\mathcal{Z}_N^*$ and $(\mathcal{B}^{-1}\mathcal{N})^T$). The notation uses calligraphic letters ($\mathcal{J}, \mathcal{X}, \mathcal{Z}, \mathcal{B}, \mathcal{N}$) to represent matrices and vectors. A red curved arrow points to the bottom right entry of the top table and is annotated "negative transpose," indicating that the matrix is obtained by taking the negative transpose of another matrix. The superscript * denotes optimal values. The context of the slide is clearly within the simplex method, specifically showing the matrix operations involved in updating the solution and objective function during iterations. The two tables likely represent the primal and dual simplex methods, respectively, highlighting the relationship between them.

Figures



Slide 12

Content

Simplex Iterations in Matrix Form

$\begin{aligned} \max \quad & \mathcal{J}(x) = \mathcal{J}^* - (\mathcal{Z}_N^*)^T x_N \\ \text{s.t.} \quad & x_B = x_B^* - (B^{-1}N)x_N \\ & x_N, x_B \geq 0 \end{aligned}$	$\begin{aligned} \max \quad & -\mathcal{J}(z) = -\mathcal{J}^* - (x_B^*)^T z_B \\ \text{s.t.} \quad & z_N = (z_N^*)^T + (B^{-1}N)^T z_B \\ & z_B, z_N \geq 0 \end{aligned}$
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$J^* = c_B^T B^{-1}b, \quad \mathcal{Z}_N^* = (B^{-1}N)^T c_B - c_N, \quad x_B^* = B^{-1}b$

Description

This slide, from a Linear Programming course, presents the primal and dual simplex methods in matrix form. The top box, labeled (P), shows the primal simplex method. It maximizes the objective function $\mathcal{J}(x)$, which is expressed as the optimal objective function value $\mathcal{J}^*$ minus a term involving the optimal values of the dual variables associated with non-basic variables, $\mathcal{Z}_N^*$, and the non-basic variables x_N . The constraints show the basic variables x_B expressed in terms of the optimal basic variable values x_B^* , the inverse of the basic variable matrix B^{-1} , the non-basic variable matrix N , and the non-basic variables x_N . The bottom box, labeled (D), shows the dual simplex method. It maximizes the negative of the dual objective function $-\mathcal{J}(z)$, expressed similarly to the primal form, using the optimal dual objective function value $-\mathcal{J}^*$, the optimal basic variable values x_B^* , and the dual variables z_B . The constraints express the dual non-basic variables z_N in terms of their optimal values z_N^* , the transpose of the inverse of the basic variable matrix $(B^{-1})^T$, the transpose of the non-basic variable matrix N^T , and the dual variables z_B . Below the boxes, the optimal objective function value $\mathcal{J}^*$, the optimal values of the dual variables associated with non-basic variables $\mathcal{Z}_N^*$, and the optimal basic variable values x_B^* are defined in terms of the cost vector c_B , the basic variable matrix B , the non-basic variable matrix N , and the right-hand side vector b . The notation clearly indicates the use of matrix operations within the simplex method. The superscript * denotes optimal values. The labels (P) and (D) likely stand for "Primal" and "Dual," respectively.

Figures

<u>Simplex Iterations in M</u> $\max \mathcal{J}(x) =$	<u>Simplex Iterations in M</u> $) = \mathcal{J}^* - (\mathcal{Z}_N^*)^T x$	x_N
(a) (P) Box containing primal simplex iteration equations.	(b) (D) Box containing dual simplex iteration equations.	(c) Equations for J^*, $\mathcal{Z}_N^*$, and x_B^*.

Slide 13

Content

Simplex Iterations in Matrix Form

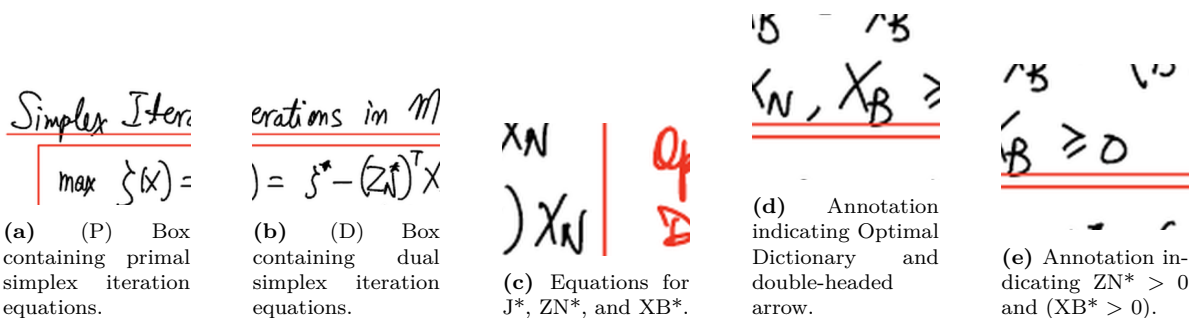
$\begin{aligned} \max \quad & \mathcal{J}(x) = \mathcal{J}^* - (\mathcal{Z}_N^*)^T \mathcal{X}_N \\ \text{s.t.} \quad & \mathcal{X}_B = \mathcal{X}_B^* - (\mathcal{B}^{-1}\mathcal{N})\mathcal{X}_N \\ & \mathcal{X}_N, \mathcal{X}_B \geq 0 \end{aligned}$	$\begin{aligned} \max \quad & -\mathcal{J}(z) = -\mathcal{J}^* - (\mathcal{X}_B^*)^T \mathcal{Z}_B \\ \text{s.t.} \quad & \mathcal{Z}_N = (\mathcal{Z}_N^*) + (\mathcal{B}^{-1}\mathcal{N})^T \mathcal{Z}_B \\ & \mathcal{Z}_B, \mathcal{Z}_N \geq 0 \end{aligned}$
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$\mathcal{J}^* = \mathcal{C}_B^T \mathcal{B}^{-1}b, \quad \mathcal{Z}_N^* = (\mathcal{B}^{-1}\mathcal{N})^T \mathcal{C}_B - \mathcal{C}_N, \quad \mathcal{X}_B^* = \mathcal{B}^{-1}b$

Description

This slide from a Linear Programming course presents the primal and dual simplex methods in matrix form, focusing on the optimal dictionary. The top box, labeled (P), shows the primal simplex method. It maximizes the objective function $\mathcal{J}(x)$, expressed as the optimal objective function value $\mathcal{J}^*$ minus a term involving the optimal dual variables associated with non-basic variables ($\mathcal{Z}_N^*$) and the non-basic variables ($\mathcal{X}_N$). The constraints express the basic variables ($\mathcal{X}_B$) in terms of their optimal values ($\mathcal{X}_B^*$), the inverse of the basic variable matrix ($\mathcal{B}^{-1}$), the non-basic variable matrix ($\mathcal{N}$), and the non-basic variables ($\mathcal{X}_N$). The bottom box, labeled (D), shows the dual simplex method. It maximizes the negative of the dual objective function ($-\mathcal{J}(z)$), expressed similarly to the primal form, using the optimal dual objective function value ($-\mathcal{J}^*$), the optimal basic variable values ($\mathcal{X}_B^*$), and the dual variables ($\mathcal{Z}_B$). The constraints express the dual non-basic variables ($\mathcal{Z}_N$) in terms of their optimal values ($\mathcal{Z}_N^*$), the transpose of the inverse of the basic variable matrix ($(\mathcal{B}^{-1})^T$), the transpose of the non-basic variable matrix ($\mathcal{N}^T$), and the dual variables ($\mathcal{Z}_B$). Below the boxes, the optimal objective function value ($\mathcal{J}^*$), the optimal values of the dual variables associated with non-basic variables ($\mathcal{Z}_N^*$), and the optimal basic variable values ($\mathcal{X}_B^*$) are defined in terms of the cost vector ($\mathcal{C}_B$), the basic variable matrix ($\mathcal{B}$), the non-basic variable matrix ($\mathcal{N}$), and the right-hand side vector (b). A double-headed arrow connects the two boxes, labeled "Optimal Dictionary," indicating the relationship between the primal and dual optimal solutions. Additional annotations indicate the conditions for optimality: $\mathcal{Z}_N^* > 0$ and $\mathcal{X}_B^* \geq 0$. The notation clearly indicates the use of matrix operations within the simplex method. The superscript $*$ denotes optimal values. The labels (P) and (D) likely stand for "Primal" and "Dual," respectively.

Figures



Slide 14

Content

Complementary Variables

$$(x_1, x_2, \dots, x_n, \quad x_{n+1}, \dots, x_{n+m}) \uparrow (z_1, z_2, \dots, z_n, \quad z_{n+1}, \dots, z_{n+m}) \textcircled{1} \quad x_j \sim z_j \quad (x_j z_j = 0) \quad w_i \sim y_i \quad (w_i y_i = 0) \textcircled{2} \quad x_j$$

This slide from a Linear Programming course describes complementary variables. It introduces two sets of variables: $(x_1, x_2, \dots, x_n, x_{n+1}, \dots, x_{n+m})$ and $(z_1, z_2, \dots, z_n, z_{n+1}, \dots, z_{n+m})$. These are paired as complementary variables, indicated by the red upward arrows connecting them. The notation $x_j \sim z_j$ signifies that x_j and z_j are complementary, and they satisfy the complementarity condition $x_j z_j = 0$. Similarly, $w_i \sim y_i$ with $w_i y_i = 0$. The slide further explains the relationship between these variables during simplex iterations. Point (2) states that when a variable x_j leaves or enters the basis, its complementary variable z_j enters or leaves, respectively. The same relationship holds for w_i and y_i . Point (3) summarizes the relationship between the sets of variables: $x_N \sim z_N$ and $x_B \sim z_B$, where x_N and x_B represent non-basic and basic variables, respectively, and similarly for z_N and z_B . Underlining is used to emphasize key relationships and conditions. The context of the slide is within the simplex method, specifically highlighting the complementary slackness conditions and how they are manifested in the algorithm's iterations.

Figures

(a) $(x_1, x_2, \dots, x_n, x_{n+1}, \dots, x_{n+m})$

(b) $(z_1, z_2, \dots, z_n, z_{n+1}, \dots, z_{n+m})$

(c) $(x_j z_j = 0)$

(d) $(w_i y_i = 0)$

(e) $(x_j \text{ leaves/enters})$

Slide 15

Content

[V] p.95

①

maximize $4x_1 + 3x_2$

subject to $x_1 - x_2 \leq 1$

$2x_1 - x_2 \leq 3$

$x_2 \leq 5$

$x_1, x_2 \geq 0$

The matrix A is given by $\begin{bmatrix} 1 & -1 & 1 & 0 & 0 \\ 2 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$

(Note that some zeros have not been shown.)

The initial sets of basic and nonbasic indices are

$B = \{3, 4, 5\}$ and $N = \{1, 2\}$.

Corresponding to these sets, we have the submatrices of A :

$B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad N = \begin{bmatrix} 1 & -1 \\ 2 & -1 \\ 0 & 1 \end{bmatrix} \quad x_1 - x_2 + x_3 = 1$

$2x_1 - x_2 + x_4 = 3$

$x_2 + x_5 = 5$

Description

This slide, from a Linear Programming course, presents a linear programming problem and its matrix representation. The problem is to maximize the objective function $4x_1 + 3x_2$ subject to the constraints $x_1 - x_2 \leq 1$, $2x_1 - x_2 \leq 3$, $x_2 \leq 5$, and $x_1, x_2 \geq 0$. The matrix A is given, representing the coefficients of the constraints. Some zero entries are omitted for brevity. The initial sets of basic and nonbasic indices are identified as $B = \{3, 4, 5\}$ and $N = \{1, 2\}$, respectively. The submatrices B and N are extracted from A corresponding to these index sets. The slide also shows the constraints rewritten in standard form using slack variables x_3, x_4, x_5 . The circled '0' likely indicates a starting point or iteration number. The notation suggests the problem is being set up for solution using the simplex method.

Figures

(a) (0)

(b) [V] p.95

(c) (=1)

(d) (=3)

(e) (=5)

The matrix A is given

$$\begin{bmatrix} 1 & -1 & 1 & 0 & 0 \\ 2 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

we have no

$B = \{3, 4, 5\}$

Slide 16

Content

Current Dictionary

maximize $\zeta = 4x_1 + 3x_2$
subject to $w_1 = 1x_1 - 1x_2 - x_1$
 $w_2 = 3 - 2x_1 - 1x_2$
 $w_3 = 5 - 0x_1 - 1x_2$
 $x_1, x_2, w_1, w_2, w_3 \geq 0$

Description

This slide from a Linear Programming course displays a linear program in the form of a "current dictionary" and its graphical representation. The linear program aims to maximize the objective function $\zeta = 4x_1 + 3x_2$ subject to the constraints $w_1 = 1x_1 - 1x_2 - x_1$, $w_2 = 3 - 2x_1 - 1x_2$, and $w_3 = 5 - 0x_1 - 1x_2$, with all variables non-negative. The dictionary is presented in a tabular format, showing the objective function and the constraints expressed in terms of basic variables (w_1, w_2, w_3) and non-basic variables (x_1, x_2). The graphical representation shows the feasible region defined by the constraints, with the axes labeled x_1 and x_2 , and the constraint lines labeled $w_1 = 0$, $w_2 = 0$, and $w_3 = 0$. The buttons "Optimal," "Infeasible," and "Unbounded" suggest that the student is using software to solve the linear program and determine the nature of the solution. The "Pick a judge" field is likely a playful addition to the software interface.

Figures



(a) Current Dictionary Table



(b) Graphical Representation of the Linear Program

Slide 17

Content

$$x_1 \geq 1$$

$$x_1 \geq 3x_2$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

$$\text{maximize } 4x_1 + 3x_2$$

$$\text{subject to } x_1 - x_2 \leq 1$$

$$2x_1 - x_2 \leq 3$$

$$x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

The matrix A is given by

$$\begin{bmatrix} 1 & -1 & 1 & 0 & 0 \\ 2 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(Note that some zeros have not been shown.)

The initial sets of basic and nonbasic indices are

$$B = \{3, 4, 5\} \quad \text{and} \quad N = \{1, 2\}.$$

Corresponding to these sets, we have the submatrices of A :

$$B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad N = \begin{bmatrix} 1 & -1 \\ 2 & -1 \\ 0 & 1 \end{bmatrix}$$

$$x_1 - x_2 + x_3 = 1$$

$$2x_1 - x_2 + x_4 = 3$$

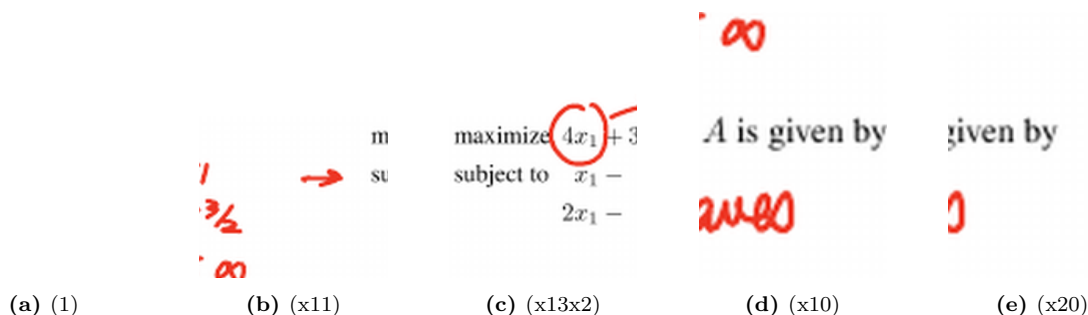
$$x_2 + x_5 = 5$$

x_3 leaves

Description

This slide from a Linear Programming course shows the initial setup for solving a linear program using the simplex method. The problem is to maximize $4x_1 + 3x_2$ subject to $x_1 - x_2 \leq 1$, $2x_1 - x_2 \leq 3$, $x_2 \leq 5$, and $x_1, x_2 \geq 0$. The initial inequalities are shown, along with the matrix A representing the constraint coefficients. Slack variables x_3, x_4, x_5 are introduced to convert inequalities to equalities. The initial basic ($B = \{3, 4, 5\}$) and non-basic ($N = \{1, 2\}$) variable sets are identified, along with the corresponding submatrices of A . The equations representing the constraints in standard form are also shown. The circled "1" likely indicates the first iteration or step in the simplex algorithm. The annotation " x_3 leaves" indicates that the variable x_3 will leave the basis in the next iteration, and " x_1 enters, largest coeff." indicates that x_1 will enter the basis because it has the largest coefficient in the objective function. The handwritten notes on the left seem to be preliminary calculations or observations about the problem's constraints.

Figures



Slide 18

Content

Step 9. The new sets of basic and nonbasic indices are

$$B = \{1, 4, 5\} \text{ and } N = \{3, 2\}.$$

Corresponding to these sets, we have the new basic and nonbasic submatrices of A ,

$$B = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ 0 & & 1 \end{bmatrix} \quad N = \begin{bmatrix} 1 & -1 \\ 0 & -1 \\ 0 & 1 \end{bmatrix},$$

and the new basic primal variables and nonbasic dual variables:

$$x_B^* = \begin{bmatrix} x_1^* \\ x_4^* \\ x_5^* \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} \quad z_N^* = \begin{bmatrix} z_3^* \\ z_2^* \end{bmatrix} = \begin{bmatrix} 4 \\ -7 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ maximize } \zeta = 4 + (-4)w_3 + 7x_2$$

subject to $x_1 = 1 - w_3 - x_2$

$$w_1 = 1 - 2w_3 - x_2$$

$$w_2 = 5 - x_2$$

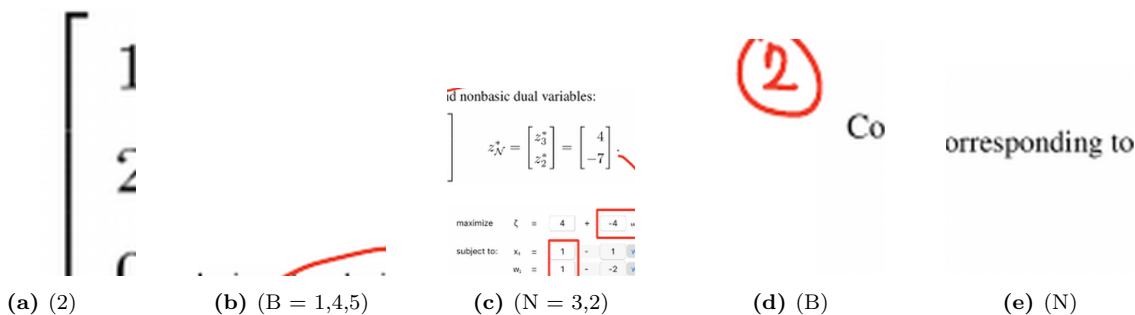
$x_1, x_2, w_1, w_2, w_3 \geq 0$ x_2 enters

x_4 leaves

Description

This slide, from a Linear Programming course, shows the 9th step of the simplex method. The new basic ($B = \{1, 4, 5\}$) and nonbasic ($N = \{3, 2\}$) index sets are highlighted. The corresponding submatrices B and N are extracted from the constraint matrix A . The basic primal variables (x_1, x_4, x_5) and nonbasic dual variables (z_3, z_2) are shown with their values. A simplex tableau is presented, showing the updated objective function and constraints after the pivot operation. The objective function is expressed as $\zeta = 4 + (-4)w_3 + 7x_2$, indicating that x_2 is the entering variable and w_3 is a nonbasic variable. The constraints are rewritten in terms of the new basic variables. The annotations "x2 enters" and "x4 leaves" indicate that x_2 will enter the basis and x_4 will leave the basis in this iteration. The matrix A at the bottom left is likely the original constraint matrix. The circled '2' likely indicates the second iteration or step in the simplex algorithm.

Figures



Slide 19

Content

Step 9. The new sets of basic and nonbasic indices are

$$B = \{1, 2, 5\} \text{ and } N = \{3, 4\}.$$

Corresponding to these sets, we have the new basic and nonbasic submatrices of A ,

$$B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix},$$

and the new basic primal variables and nonbasic dual variables:

$$x_B^* = \begin{bmatrix} x_1^* \\ x_2^* \\ x_5^* \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} \quad z_N^* = \begin{bmatrix} z_3^* \\ z_4^* \end{bmatrix} = \begin{bmatrix} -10 \\ 7 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ Current Dictionary}$$

$$\text{maximize } \zeta = 11 + 10w_3 - 7w_4$$

$$\text{subject to } x_1 = 2 - w_3 - w_4$$

$$x_2 = 1 - 2w_3 + w_4$$

$$w_1 = 4 + 2w_3 - w_4$$

$$x_1, x_2, w_1, w_3, w_4 \geq 0$$

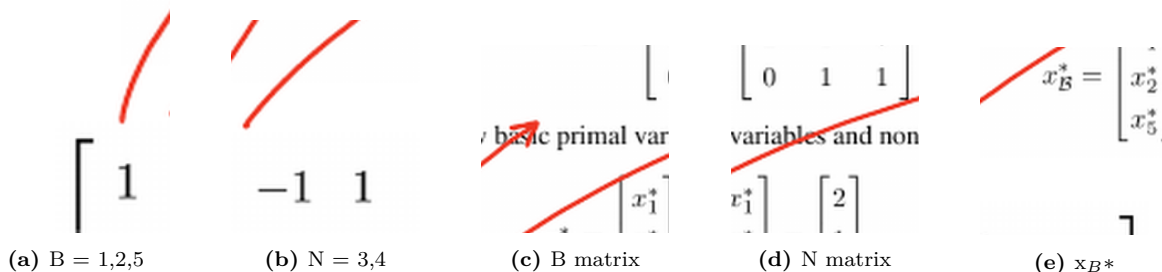
x_3 enters

x_5 leaves

Description

This slide presents the ninth step in solving a linear program using the simplex method. The new basic variables are x_1 , x_2 , and x_5 , forming the set $B = \{1, 2, 5\}$, while the non-basic variables are x_3 and x_4 , forming the set $N = \{3, 4\}$. The corresponding submatrices B and N are shown, extracted from the constraint matrix A . The values of the basic primal variables ($x_1^* = 2$, $x_2^* = 1$, $x_5^* = 4$) and the non-basic dual variables ($z_3^* = -10$, $z_4^* = 7$) are given. A "Current Dictionary" table is displayed, showing the updated objective function ($\zeta = 11 + 10w_3 - 7w_4$) and constraints expressed in terms of the current basic and non-basic variables. The annotations " x_3 enters" and " x_5 leaves" indicate that x_3 will enter the basis and x_5 will leave the basis in the next iteration of the simplex method. The handwritten notes on the left seem to be preliminary calculations or observations about the problem's constraints. The context of the course (Linear Programming) is crucial for understanding the simplex method and its application in solving optimization problems.

Figures



Slide 20

Content

Step 9. The new sets of basic and nonbasic indices are

$$B = \{1, 2, 3\} \text{ and } N = \{5, 4\}.$$

Corresponding to these sets, we have the new basic and nonbasic submatrices of A ,

$$B = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad N = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix},$$

and the new basic primal variables and nonbasic dual variables:

$$x_B^* = \begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 2 \end{bmatrix} \quad z_N^* = \begin{bmatrix} z_5^* \\ z_4^* \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \text{ Current Dictionary}$$

$$\text{maximize } \zeta = 31 - 5w_5 - 2w_4$$

$$\text{subject to } x_1 = 4 - \frac{1}{2}w_5 - \frac{1}{2}w_4$$

$$x_2 = 5 - w_5$$

$$x_3 = 2 - \frac{1}{2}w_5 + \frac{1}{2}w_4$$

$$x_1, x_2, x_3, w_4, w_5 \geq 0$$

$$\text{Optimal as } z_N^* \geq 0 \text{ and } x_B^* \geq 0$$

Description

This slide shows the 9th step in solving a linear program using the simplex method. The new basic variables are x_1 , x_2 , and x_3 , forming the set $B = \{1, 2, 3\}$, while the non-basic variables are x_5 and x_4 , forming the set $N = \{5, 4\}$. The corresponding submatrices B and N are shown, extracted from the constraint matrix A . The values of the basic primal variables ($x_1^* = 4$, $x_2^* = 5$, $x_3^* = 2$) and the non-basic dual variables ($z_5^* = 5$, $z_4^* = 2$) are given. A "Current Dictionary" table is displayed, showing the updated objective function ($\zeta = 31 - 5w_5 - 2w_4$) and constraints expressed in terms of the current basic and non-basic variables. The condition for optimality is stated: $z_N^* \geq 0$ and $x_B^* \geq 0$, indicating that the optimal solution has been reached. The handwritten "Opt." likely marks this as the optimal solution. The original constraint matrix A is also shown at the bottom left. The context of the course (Linear Programming) is crucial for understanding the simplex method and its application in solving optimization problems.

Figures



(a) Opt.

new sets of basic a

$$B = \{1, 2, 3\}$$

(b) B matrix

and nonbasic indic

$$N = \{5, 4\}$$

(c) N matrix

to these sets, we

$$B = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

(d) x_B^*

have the new ba

(e) z_N^*

Slide 21

Content

Current Dictionary

$$\text{maximize } \zeta = 31 - 5w_5 - 2w_4$$

$$\text{subject to } x_1 = 4 - \frac{1}{2}w_5 - \frac{1}{2}w_4$$

$$x_2 = 5 - w_5$$

$$x_3 = 2 - \frac{1}{2}w_5 + \frac{1}{2}w_4$$

$$x_1, x_2, x_3, w_4, w_5 \geq 0$$

$$\zeta^* = 31 \quad (= 4x_1^* + 3x_2^* = 4(4) + 3(5))$$

Description

This slide, the final slide (21 of 21), displays the optimal solution obtained through the simplex method. The "Current Dictionary" table shows the optimal values of the objective function ($\zeta^* = 31$) and the basic variables ($x_1^* = 4$, $x_2^* = 5$, $x_3^* = 2$), with all non-basic variables (w_4, w_5) equal to zero. The handwritten calculation $\zeta^* = 31 = 4x_1^* + 3x_2^* = 4(4) + 3(5)$ verifies the optimal objective function value using the optimal values of x_1 and x_2 . A graphical representation of the simplex method is shown, illustrating the path taken to reach the optimal solution. The arrows indicate the movement through the feasible region, with each arrow representing a pivot step in the simplex algorithm. The labels "3 opt" and the coordinates on the graph visually confirm the optimal solution found in the table. The buttons "Optimal," "Infeasible," and "Unbounded" suggest that the software used allows for checking the nature of the solution. The context of the course (Linear Programming) is crucial for understanding the simplex method and its application in solving optimization problems. The slide summarizes the entire process, showing both the tabular and graphical representations of the solution.

Figures

(a) (Current Dictionary Table)

Current Dictionary					
ζ	$=$	31	$+$	-5	w_5
x_1	$=$	4	$+$	1/2	w_5
x_2	$=$	5	$+$	1	w_5
x_3	$=$	2	$+$	1/2	w_5

(b) (Graphical Representation of Simplex Method)

(c) (Handwritten Calculation of Optimal Value)